

Exploratory Data Analysis on Radiology Dataset

Dataset: This dataset contains radiology report texts extracted from chest imaging exams, along with associated clinical metadata. It is intended for natural language processing (NLP) and exploratory data analysis (EDA) to identify key patterns in medical imaging documentation.

Dataset Link: [Link](#)

Dimensions:

- Total no.of rows or observations = 954
- Total no.of columns = 6

Column Name	Data Type	Description
ReportText	Text / Object	Full textual radiology report.
findings	Text / Object	Radiologist's technical observations.
clinicaldata	Text / Object	Reason for examination (e.g., symptoms).
ExamName	Text / Object	Exam type and date.
impression	Text / Object	Final diagnosis or summary.

	ReportText	findings	clinicaldata	ExamName	impression
0	EXAM: CHEST RADIOGRAPHY EXAM DATE: 06/01/2019 ...	FINDINGS: Lungs/Pleura: No focal opacities evi...	CLINICAL HISTORY: Cough. \n\n	EXAM: CHEST RADIOGRAPHY EXAM DATE: 06/01/2019 ...	IMPRESSION: Normal 2-view chest radiography.
1	EXAM: CHEST RADIOGRAPHY EXAM DATE: 05/23/2020 ...	FINDINGS: Lungs/Pleura: No focal opacities evi...	CLINICAL HISTORY: CHEST PAIN. \n\n	EXAM: CHEST RADIOGRAPHY EXAM DATE: 05/23/2020 ...	IMPRESSION: No acute cardiopulmonary abnormali...
2	EXAM: CHEST RADIOGRAPHY EXAM DATE: 12/13/2019 ...	FINDINGS: Lungs/Pleura: No focal opacities evi...	CLINICAL HISTORY: CHEST PAIN. \n\n	EXAM: CHEST RADIOGRAPHY EXAM DATE: 12/13/2019 ...	IMPRESSION: No acute cardiopulmonary process.
3	Exam: - CHEST-PORTABLE History: Chest pain Com...	Findings: Heart size appears normal. Lungs cle...	History: Chest pain \n\n	Exam: - CHEST-PORTABLE\n\nComparison: None	Impression: Lungs clear
4	EXAM: CHEST RADIOGRAPHY EXAM DATE: 06/17/2021 ...	FINDINGS: Lungs/Pleura: No focal opacities evi...	CLINICAL HISTORY: CHEST PAIN, SHORTNESS OF BRE...	EXAM: CHEST RADIOGRAPHY EXAM DATE: 06/17/2021 ...	IMPRESSION: Normal single view chest.

Null Values:

- All fields contain non-null values except `clinicaldata` field.
- It has 4 missing values as below and hence those rows are dropped.

```
data.isnull().sum()

ReportText      0
findings        0
clinicaldata     4
ExamName        0
impression      0
dtype: int64
```

Duplicates:

- There are no duplicate records, but it is observed that all fields have repeated values which are like templates.
- Below are the most repeated values for each field/column in our Dataset.

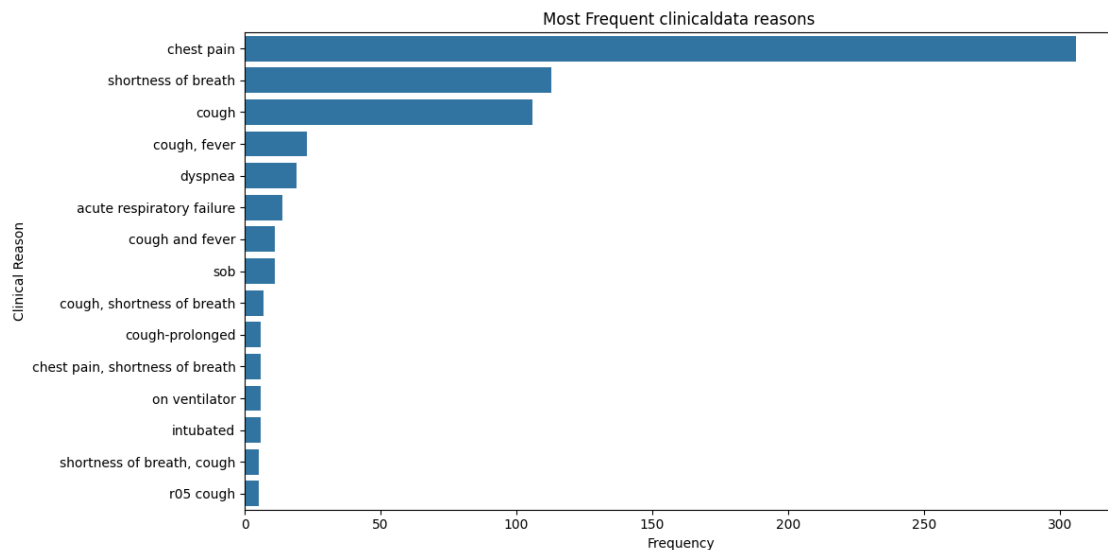
	Most Frequent Value	Frequency
ReportText	EXAM: CHEST RADIOGRAPHY EXAM DATE: 06/01/2019 08:30 PM. CLINICAL HISTORY: Cough. COMPARISON: None. TECHNIQUE: 2 views. FINDINGS: Lungs/Pleura: No focal opacities evident. No pleural effusion. No pneumothorax. Normal volumes. Mediastinum: Heart and mediastinal contours are unremarkable. Other: None. IMPRESSION: Normal 2-view chest radiography.	1
findings	FINDINGS: Lungs/Pleura: No focal opacities evident. No pleural effusion. No pneumothorax. Mediastinum: Within exam limitations, the cardiomedastinal contour is normal. Other: None. \n\n	174
clinicaldata	CLINICAL HISTORY: Chest pain. \n\n	84
ExamName	EXAM: CHEST 2 VIEW\n\nCOMPARISON: None.	35
impression	IMPRESSION: Normal single view chest.	160

Preprocessing Clinicaldata Field:

- To identify the most common reason or symptoms, I have preprocessed clinical data field like lower casing text, splitting, stripping extra space, and removing special characters.

```
data['clinical_reason'] = data['clinicaldata'].str.split(':').str[-1].str.lower().str.strip()
data['clinical_reason'] = data['clinical_reason'].str.replace('.', '')
reason_counts = data['clinical_reason'].value_counts().head(15)
data.drop("clinical_reason",axis=1,inplace=True)
```

- **"Chest pain"** is the most frequent clinical reason/symptoms for radiology exams followed by **"shortness of breath"** and **"cough"**.



Analyzing Text Statistics: Text statistics visualizations include:

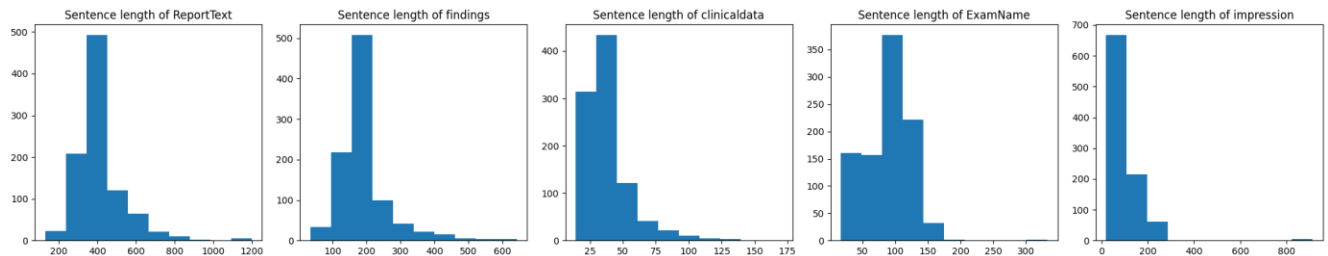
- sentence length analysis
- word frequency analysis
- average word length analysis

Sentence level - Text Analysis:

The ReportText field is the longest and most detailed, with an average length of 405 characters and no duplicates, while clinicaldata shows high duplication, indicating limited variation in symptoms for radiology.

	Min Length	Max Length	Avg Length	Unique Entries	Duplicates count
ReportText	128.0	1197.0	405.240000	950.0	0.0
findings	34.0	642.0	193.044211	406.0	544.0
clinicaldata	14.0	171.0	37.565263	367.0	583.0
ExamName	17.0	333.0	90.680000	732.0	218.0
impression	19.0	914.0	88.806316	486.0	464.0

Sentence length analysis of each field:

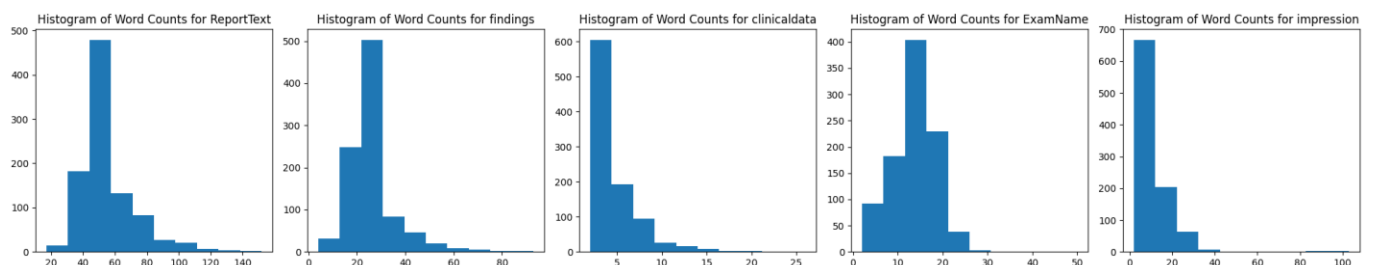


These histograms show the distribution of sentence lengths across five fields in our dataset:

- The `ReportText` field appears to contain the full narrative report, as expected. With most of sentences having length around 350-450 characters.
- Whereas `clinicaldata` and `ExamName` fields are much shorter, typically under 100-150 characters, suggesting they contain concise/precise content.
- `impression` shows a highly skewed distribution indicating it has shorter sentence lengths i.e., below 150 characters.
- All fields exhibit right-skewed distributions, with mostly shorter sentences and few longer ones.

Word level - Text Analysis

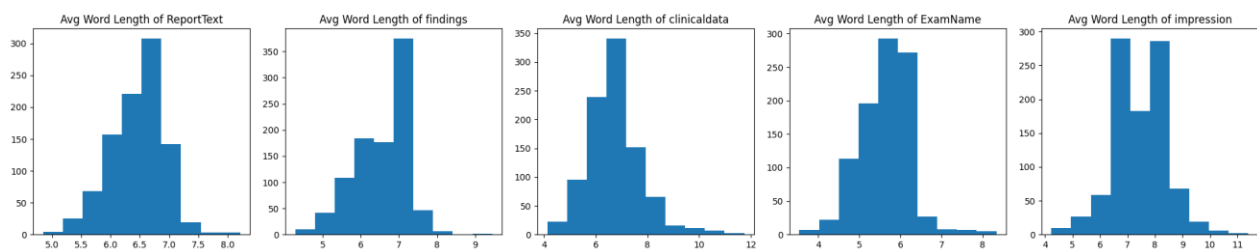
Word count (row-level) for each field in the Dataset



These histograms show the distribution of word count across five fields in our dataset:

- Similar to sentence length observations, `ReportText` and `findings` contain higher word counts, typically ranging between 20 - 80 words, indicating more detailed content.
- `clinicaldata` and `ExamName` fields have much lower word counts, mostly under 20 words, suggesting they hold brief or structured inputs whereas `impression` is mostly very short.
- Similar to sentence length, here also all fields exhibit right-skewed distributions, with mostly shorter entries with few longer ones.

Average word length (row-level) in each field:

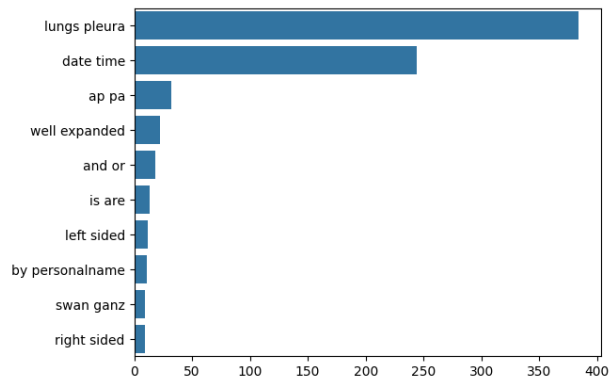


- All fields show average word lengths concentrated between 5 to 8 characters, indicating consistency in vocabulary complexity across fields.
- `clinicaldata` and `impression` show a slightly wider spread, suggesting more variability in the types of words used.
- `ExamName` has slightly shorter average word lengths, likely due to frequent use of standard, concise medical terminology.

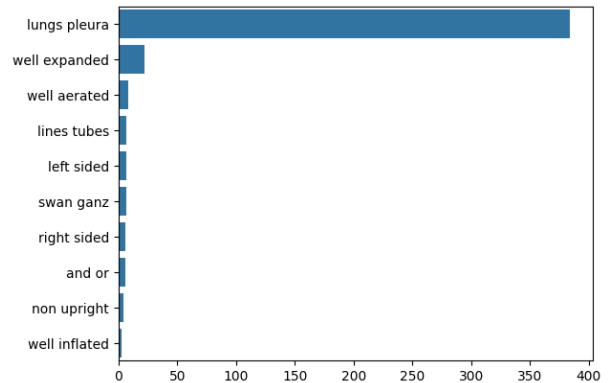
Bi-gram Analysis:

For Bi-gram analysis, I have removed stop words for better results. Analysis revealed common medical phrase pairs in each field lungs pleura, well expanded, well inflated

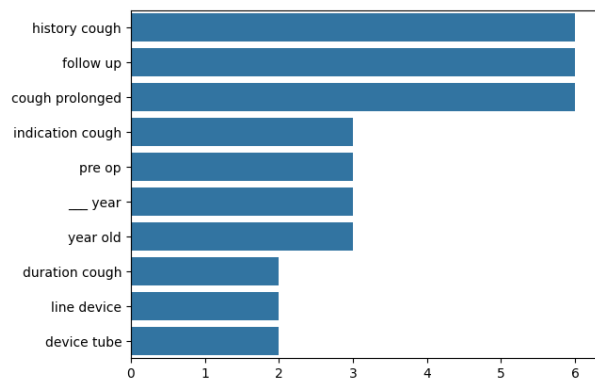
ReportText:



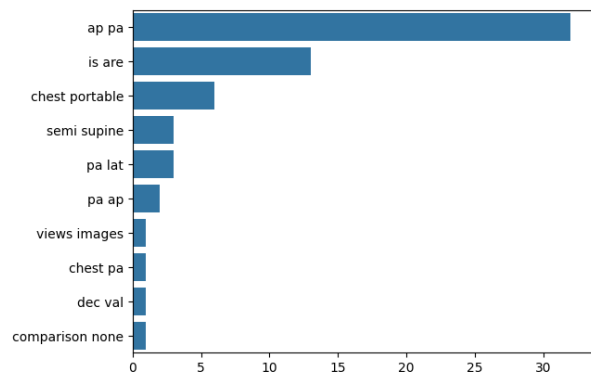
findings:



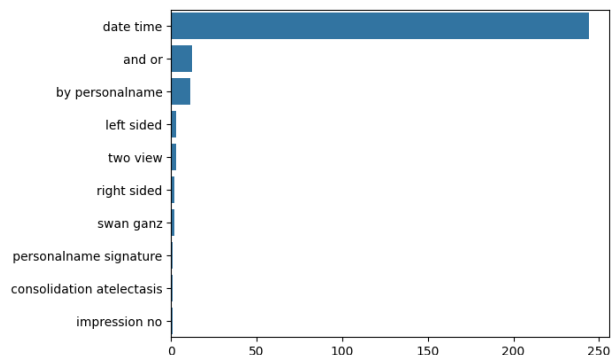
Clinical data:



ExamName:

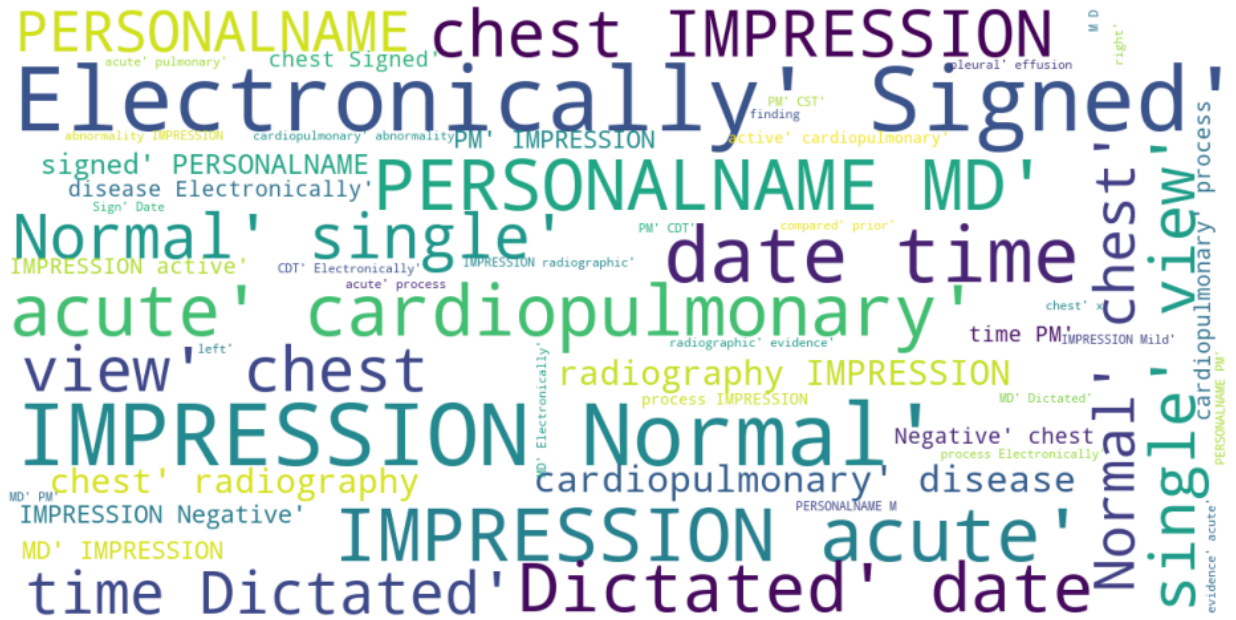


Impressions:



- **Clinicaldata word cloud** shows common symptoms like "chest pain", "shortness of breath", "cough", "History Cough".

- **Impressions word cloud** shows frequent words such as "normal", "negative", "acute", suggesting common diagnoses.



Key Findings of EDA:

- “Chest pain” is the most common clinical reason for imaging, followed by “shortness of breath” and cough”, as identified through preprocessing of the clinicaldata field.
- *findings* is the most detailed fields with high sentence and word counts while clinicaldata, ExamName, and impression contain shorter, more structured/repeated content.
- All text fields exhibit right-skewed distributions in sentence length and word count, indicating most entries are concise with a few longer and detailed reports.
- Word clouds and Bigram analysis highlight medically relevant terms like lungs, pleura, pneumothorax in *findings*, and normal, acute, negative in *impression*.
- Average word lengths across fields range between 5 and 8 characters, reflecting consistent vocabulary complexity suitable for NLP modeling.

Observation:

- **ExamName** contains Date on which examination was done, and this can be extracted as a separate column which can be helpful in further down streaming task